

Jagadeesh Arigala

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Summary

AI/ML Engineer specializing in Computer Vision, deep learning, and natural language processing, with internship and project experience designing model pipelines across data preprocessing, architecture selection, training, and evaluation. Applies MLOps principles — including experiment tracking, containerization, and CI/CD — to maintain structured, reproducible development workflows.

Education

Bachelor of Technology - VIT-AP University, Amaravati *Sep 2022 – May 2026*
CSE with specialization AI&ML | GPA: 8.51 / 10.0

Intermediate - BIEAP - Narayana Junior College, Tirupati *June 2020 – May 2022*
MPC Stream | GPA: 9.59 / 10.0

10th class - BSEAP - Keshava Reddy School, Tirupati *June 2019 – May 2020*
State Board | Marks: 594/600

Experience

AI & ML Intern - Infosys Springboard | Remote, India *Nov 2024 – Jan 2025*

- Designed and trained a multi-class **CNN classifier** for **Advanced Skin Disease Diagnosis**, achieving **94% test accuracy** across **9 disease categories** on a **27,000-image dermoscopy dataset** via transfer learning and custom architecture fine-tuning.
- Structured model development workflows using **MLOps concepts** — **experiment tracking, model versioning, and Docker-based containerization** — to ensure reproducibility across multiple training iterations.

Projects

Plant Brain — AI for Industrial Knowledge Intelligence *Live*

- Architected a **Hybrid Graph-RAG** system fusing knowledge graph traversal with vector search, auto-extracting **7 entity types** and **464 relationships** from heterogeneous industrial documents (PDF, DOCX, XLSX) via **spaCy NER**, delivering **3–5× higher retrieval precision** over standard RAG.
- Engineered and deployed a full-stack **multi-agent AI platform** on **Azure App Service** with **Docker** containerization and **GitHub Actions CI/CD**, integrating local **ONNX embeddings** (all-MiniLM-L6-v2) to eliminate external API dependency and reduce per-query embedding cost to zero.
- Delivered sub-3-second query responses across maintenance root cause analysis, compliance gap detection, and equipment history workflows — serving **3 specialized AI agents** over a **161-node knowledge graph** with **real-time WebSocket streaming**.

Diabetic Retinopathy Detection using Deep Learning

- Engineered a deep learning pipeline using **MobileNetV3Small** with **CBAM** attention in TensorFlow, achieving **89% test accuracy** on the **APTOS 2019** retinal fundus dataset for **5-grade DR severity classification**.
- Benchmarked cross-dataset generalization at **83% accuracy** on **50,000+ EyePACS images** without retraining; implemented **OpenCV CLAHE** preprocessing and **Albumentations** augmentation with class-weighted loss functions to mitigate severity grade imbalance.

Skills

Languages: Python, Java

Deep Learning & GenAI: TensorFlow, LLMs, RAG, Graph-RAG, Multi-Agent AI, Neural Networks, Transfer Learning

Machine Learning: Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn

MLOps & Tools: Docker, Git, GitHub Actions, CI/CD, Azure App Service, Vercel, FastAPI, Flask, REST APIs, WebSocket

Databases: MySQL, ChromaDB, FAISS, NetworkX (Knowledge Graph)

Certifications

- Microsoft Certified: Azure AI-900 Fundamentals** – Validation of cloud-based AI workloads and machine learning principles on Azure.
- CS50P: Introduction to Programming with Python** – Harvard University / edX. Intensive foundation in Pythonic design and data structures.
- Certified Generative AI Professional** – Oracle Cloud Infrastructure (2025). Focus on LLM architecture and deployment.